Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: Batch 11

Start date: 29th December 2025

Name: Shyam Sarath

Internship Duration: 8 Weeks

1. Project Title

Machine Learning Based Phishing Detection Website

2. Project Objective

The main objective of this project was to design and develop a Machine Learning based web application that can detect phishing URLs and protect users from cyber threats. The system aims to classify URLs as legitimate or phishing using supervised machine learning algorithms. The project demonstrates the practical application of machine learning techniques in strengthening web security and protecting user data.

c3. Project description in detail

This project involves the design and development of a Machine Learning–based web application for the detection of phishing URLs. Phishing attacks represent a significant cybersecurity threat, where malicious websites imitate legitimate platforms to obtain sensitive user information such as login credentials, financial details, and personal data. The primary objective of this system is to accurately classify URLs as either legitimate or phishing using supervised machine learning techniques.

My approach began with understanding the problem domain and studying common phishing patterns. I collected and preprocessed a labeled dataset of legitimate and phishing URLs. Relevant features such as URL length, presence of special characters, use of HTTPS, number of subdomains, redirection behavior, and domain-related attributes were extracted for analysis. After feature engineering and data cleaning, I implemented a supervised learning model using the Random Forest algorithm due to its high accuracy, robustness, and ability to reduce overfitting. The model was trained, validated, and evaluated using appropriate performance metrics to ensure reliable classification results.

The development of the project was carried out using Python for data preprocessing, feature engineering, and model training. Essential libraries such as Scikit-learn, Pandas, and NumPy were utilized to build and evaluate the Random Forest classification model efficiently. The web application interface was created using HTML, CSS, and JavaScript to ensure a simple and user-friendly experience.

The backend was implemented using the Flask framework, which enabled seamless integration of the trained machine learning model for real-time URL classification. The final system allows users to enter a URL and instantly receive a prediction indicating whether the website is legitimate or phishing.

In real-world scenarios, this solution can be integrated into web browsers, enterprise security infrastructures, or online platforms to provide an additional layer of cybersecurity protection. It can help organizations proactively detect malicious websites, reduce financial risks, and safeguard sensitive user information. Overall, the project demonstrates the effective implementation of machine learning techniques in addressing real-world cybersecurity challenges and emphasizes the role of intelligent systems in enhancing digital security.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project understanding, research on phishing attacks and dataset collection.	Understood phishing attack concepts and types. Collected phishing and legitimate URL dataset. Studied feature extraction techniques. Finalized project architecture and tools.
Week 2	Data preprocessing and feature extraction.	Cleaned and preprocessed dataset (removed null values, duplicates). Extracted important URL features (length, HTTPS, special symbols, etc.). Converted categorical data into numerical format. Prepared dataset for model training.
Week 3	Model training using Random Forest algorithm.	Implemented Random Forest algorithm. Trained model on processed dataset. Split data into training and testing sets. Generated initial accuracy results.
Week 4	Model evaluation and accuracy improvement.	Evaluated model using accuracy, precision, recall. Tuned hyperparameters to improve performance. Reduced overfitting issues. Finalized best-performing model.
Week 5	Backend development using Flask.	Created Flask backend structure. Integrated trained model (.pkl file). Developed URL prediction API logic. Tested backend with sample inputs.

Week 6	Frontend development and UI design.	Designed login page and URL scan page. Developed responsive UI using HTML, CSS, Bootstrap. Connected frontend with Flask backend. Tested UI functionality.
Week 7	Integration of model with web application and testing.	Integrated ML model with web application. Performed end-to-end testing. Fixed bugs and improved UI design.
Week 8	Final deployment, documentation, and presentation preparation.	Final deployment of project. Prepared project documentation. Created presentation slides. Submitted final report and demonstrated project.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Requirement gathering, problem identification, and initial project planning completed.	12th Jan 2026
Prototype/First Draft	Developed and tested the initial machine learning model for phishing detection.	25th Jan 2026
Mid-Term Review	Successfully integrated the ML model with the web application interface and completed mid-project evaluation.	9th Feb 2026
Final Submission	Finalized the system, prepared full documentation, and validated overall performance.	17th Feb 2026
Presentation	Delivered the complete project demonstration and explained system architecture and results.	23rd Feb 2026

5b. Project execution details

The development of PhisCheck was carried out through a systematic and well-structured process to ensure accuracy, reliability, and real-world applicability. A comprehensive dataset containing both legitimate and phishing URLs was collected from credible sources to support effective model training. The dataset was carefully cleaned to remove duplicates, inconsistencies, and irrelevant entries, ensuring high-quality input for the machine learning pipeline.

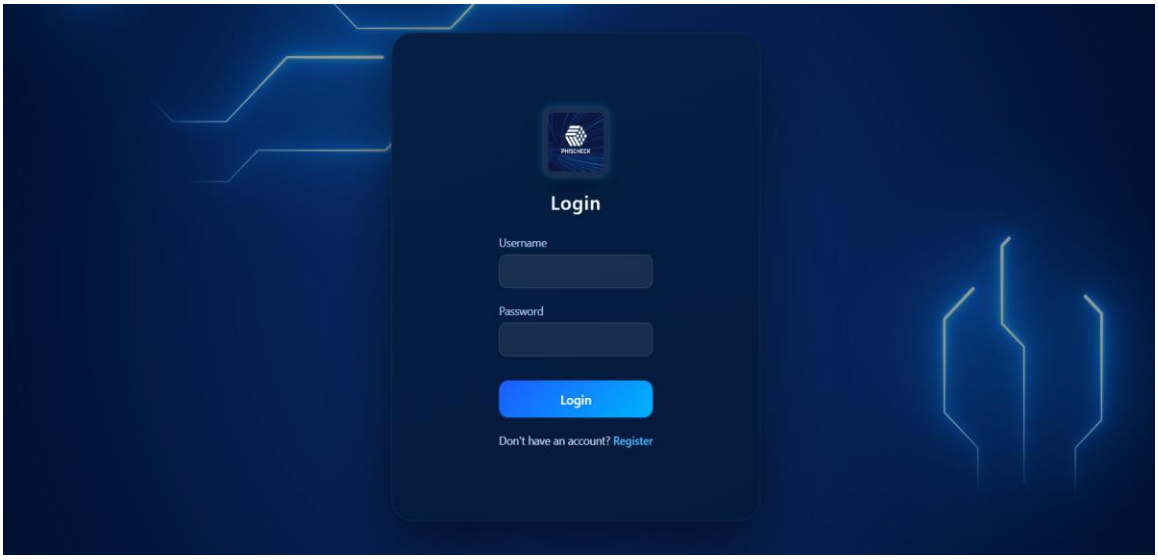
Data preprocessing and feature engineering were critical stages of the implementation. Raw URL data was transformed into structured numerical features by extracting important attributes such as URL length, presence of special characters, HTTPS usage, number of subdomains, redirection patterns, usage of IP addresses, and other domain-related characteristics. These features were selected based on their strong relevance to phishing detection.

Multiple classification algorithms were implemented and compared during the experimentation phase, including models such as Decision Tree, Logistic Regression, and Support Vector Machine. Among these, the Random Forest classifier demonstrated superior performance in terms of accuracy, precision, recall, and overall stability. It produced better results than the other models by effectively reducing overfitting and handling feature variability through its ensemble learning approach. By constructing multiple decision trees and combining their predictions using majority voting, Random Forest achieved higher generalization capability and consistent detection accuracy.

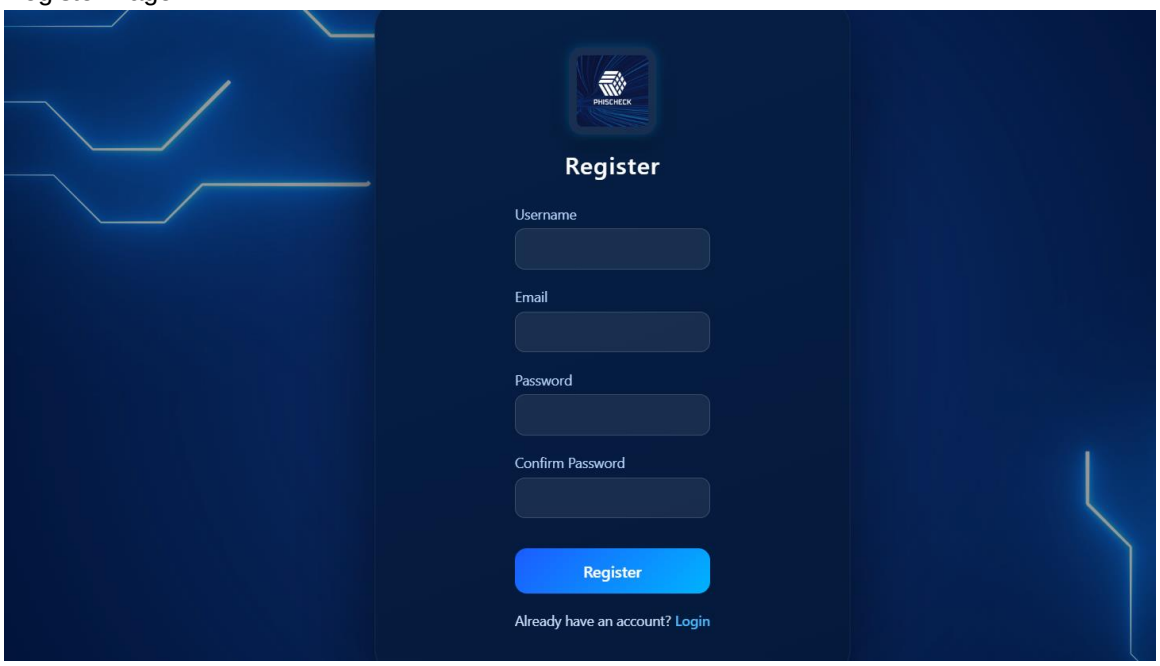
Following model selection and optimization through hyperparameter tuning, the finalized Random Forest model was integrated into the PhisCheck web application. The frontend interface was designed to provide a simple and user-friendly experience, allowing users to input URLs easily. The backend, developed using Flask, managed feature extraction and real-time prediction by connecting the deployed model. Comprehensive system testing was conducted to ensure reliability, responsiveness, and consistent classification performance prior to deployment, establishing PhisCheck as an effective phishing detection solution.

6. Snapshots / Screenshots

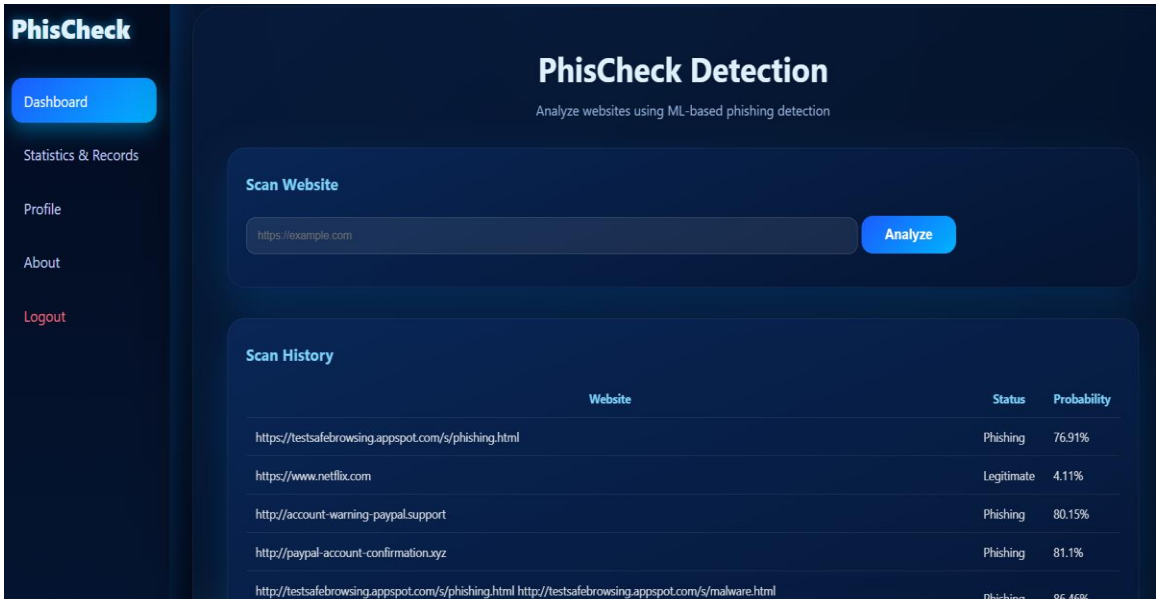
1. Login Page



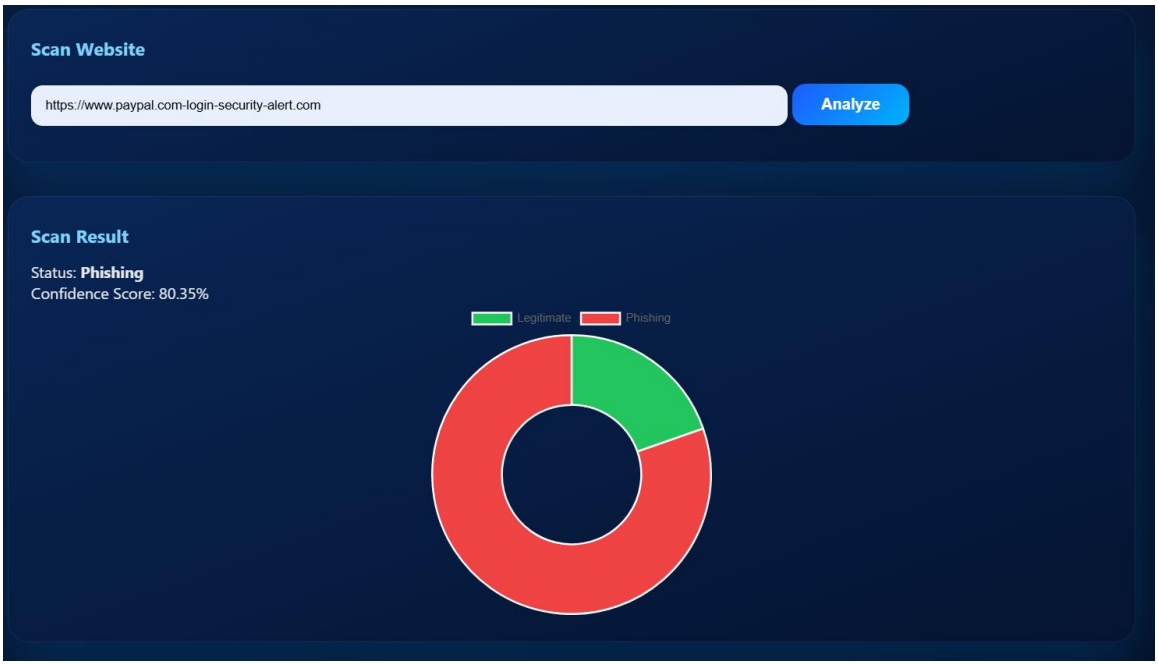
2. Register Page



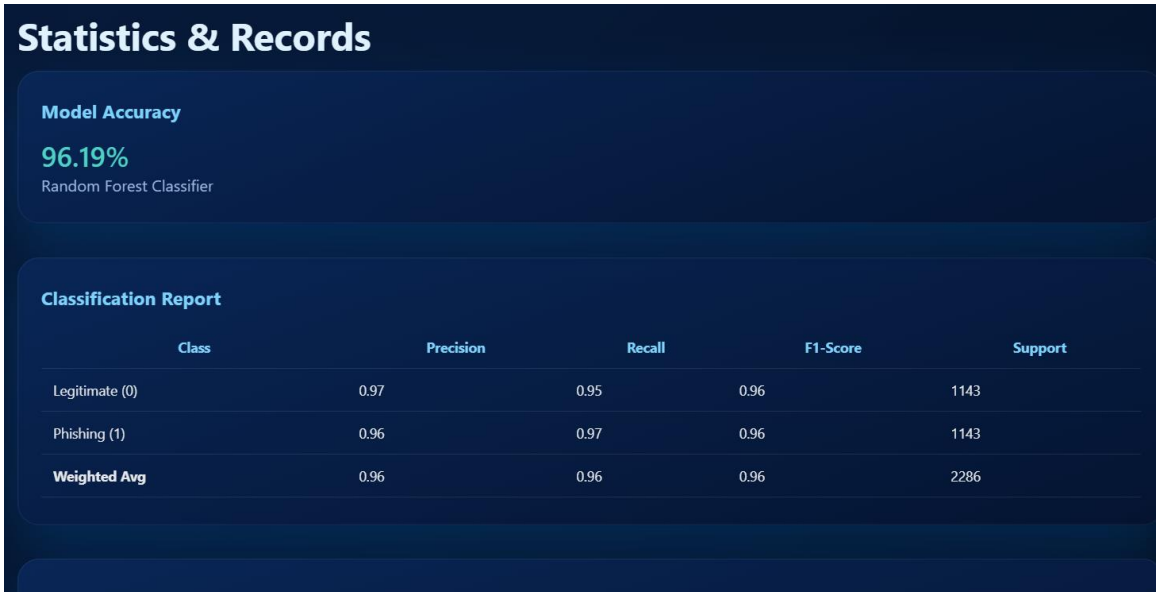
3. Dashboard



4. Analysis and confidence Score

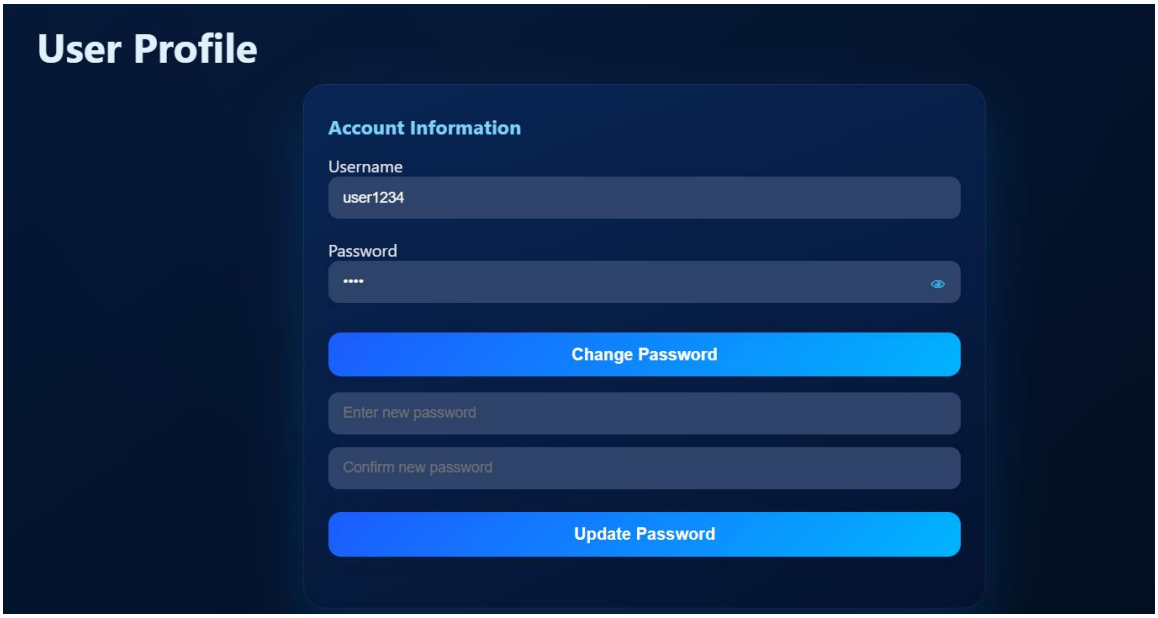


5.Stats and records

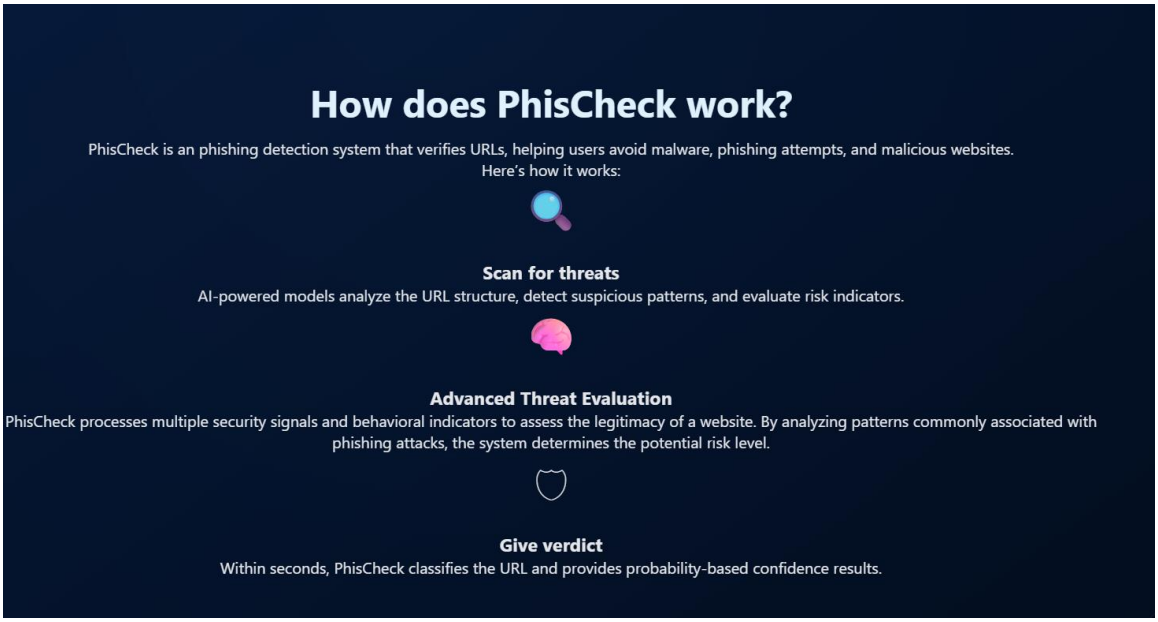


Class	Precision	Recall	F1-Score	Support
Legitimate (0)	0.97	0.95	0.96	1143
Phishing (1)	0.96	0.97	0.96	1143
Weighted Avg	0.96	0.96	0.96	2286

6. Profile Section



7. About



7. Challenges Faced

Challenges Encountered and Mitigation Strategies

During the internship, several technical, operational, and communication challenges were encountered while developing the PhisCheck phishing detection system. These challenges

provided valuable learning opportunities and contributed to both technical growth and problem-solving skills.

1. Technical Challenges

- **Imbalanced Dataset:** The dataset contained a higher proportion of legitimate URLs compared to phishing URLs, which affected model learning. This was mitigated by applying appropriate preprocessing techniques, using suitable evaluation metrics (such as precision and recall), and carefully validating model performance to ensure balanced detection capability.
- **Feature Selection and Engineering:** Identifying the most relevant URL features required experimentation and analysis. Irrelevant or low-impact features were removed after conducting correlation analysis and performance comparisons to improve model accuracy.
- **Model Optimization:** Achieving optimal performance required experimenting with multiple algorithms before selecting Random Forest. Hyperparameter tuning was performed to enhance generalization and reduce overfitting.
- **Model Integration with Backend:** Integrating the trained model with the Flask backend posed challenges related to model serialization and real-time prediction. This was resolved by properly saving the trained model and implementing structured request–response handling to ensure smooth communication between frontend and backend.

2. Operational Challenges

- **Deployment and Environment Configuration:** Managing dependencies and ensuring compatibility between development and deployment environments created initial issues. These were addressed by setting up a virtual environment, organizing project dependencies properly, and performing systematic testing before deployment.
- **Debugging and Performance Issues:** Runtime errors and delays in prediction response time required thorough debugging and code optimization to improve system stability and efficiency.

8. Learnings & Skills Acquired

The internship provided hands-on experience in developing a real-world machine learning application for cybersecurity through the PhisCheck phishing detection system. I gained practical knowledge of the complete ML workflow, including data preprocessing, feature engineering, model training, evaluation, and deployment using Python, Pandas, NumPy, and Scikit-learn.

I also developed skills in web application integration using HTML, CSS, JavaScript, and Flask to enable real-time predictions. The internship enhanced my understanding of phishing attack patterns and cybersecurity concepts. Additionally, it strengthened my problem-solving, analytical thinking, time management, and communication skills, contributing to both my technical and professional growth.

9. Testimonials from team

This internship offered meaningful real-world exposure to cybersecurity and machine learning practices. The successful development of the project demonstrated strong technical competency and a clear understanding of full-stack machine learning integration. The experience significantly enhanced practical skills and strengthened confidence in designing, developing, and deploying end-to-end ML-based web applications.

10. Conclusion

The internship was a valuable learning experience that provided practical exposure to machine learning and cybersecurity applications. Developing the PhisCheck phishing detection system allowed me to apply academic knowledge to a real-world project, strengthening my technical skills in model development and web integration. It also enhanced my problem-solving, analytical thinking, and communication abilities. This experience aligns well with my academic background and supports my long-term career goals in data-driven and analytical roles.

11. Acknowledgements

I would like to express my sincere gratitude to the organization for providing me with the opportunity to undertake this internship and gain valuable industry exposure. I am especially thankful to my mentor, Tony, for his continuous guidance, technical insights, and constructive feedback throughout the project

I would also like to thank our coordinator, Mr Santosh, and all team members who offered their assistance and encouragement during my internship journey. Their support and collaboration contributed significantly to both my professional growth and the successful completion of this project.